Physically Constrained Transfer Learning Through Shared Abundance Space for Hyperspectral Image Classification

Ying Qu, *Member, IEEE*, Razieh Kaviani Baghbaderani, *Student Member, IEEE*, Wei Li, *Senior Member, IEEE*, Lianru Gao, *Senior Member, IEEE*, Yuxiang Zhang, *Student Member, IEEE*, and Hairong Qi, *Fellow, IEEE*

***Abstract*— Hyperspectral image (HSI) classification is one of the most active research topics and has achieved promising results boosted by the recent development of deep learning. However, most state-of-the-art approaches tend to perform poorly when the training and testing images are on different domains, e.g., the source domain and target domain, respectively, due to the spectral variability caused by different acquisition conditions. Transfer learning-based methods address this problem by pre-training in the source domain and fine-tuning on the target domain. Nonetheless, a considerable amount of data on the target domain has to be labeled and nonnegligible computational resources are required to retrain the whole network. In this article, we propose a new transfer learning scheme to bridge the gap between the source and target domains by projecting the HSI data from the source and target domains into a shared abundance space based on their own physical characteristics. In this way, the domain discrepancy would be largely reduced such that the model trained on the source domain could be applied to the target domain without extra efforts for data labeling or network retraining. The proposed method is referred to as physically constrained transfer learning through shared abundance space (PCTL-SAS). Extensive experimental results demonstrate the superiority of the proposed method as compared to the state of the art. The success of this endeavor would largely facilitate the deployment of HSI classification for real-world sensing scenarios.**

***Index Terms*— Deep learning, hyperspectral image (HSI) classification, physical constraints, transfer learning.**

I. Introduction

HYPERSPECTRAL images (HSIs) collect rich spectral information from hundreds of narrow and contiguous spectral bands covering the electromagnetic spectrum from visible to near-infrared wavelengths. Because of the improved image acquisition technologies, HSI analysis has been playing a very important role in the field of remote sensing. Among the numerous advanced techniques for HSI analysis, HSI classification has become one of the most active research topics and contributes significantly in a wide area of applications, including ecological science, ecology management, precision agriculture, military applications, and geological and planetary exploration [1]–[7].

HSI classification aims to categorize each individual pixel of HSI into one of the given classes based on either the raw input data or features extracted from the raw data. Hence the classification performance relies heavily on the effectiveness of the feature extraction procedure. Traditional HSI classification approaches mainly extract hand-crafted (or engineered) shallow features [4], [8], [9]. One major limitation of these early approaches has is the limited generalization capacity when applied to complex scenarios [5]. Recently, deep learning-based approaches have been developed that automatically extract deep features from the input data. These deep features have demonstrated unique strength in HSI classification. The deep learning-based approaches can be generally categorized into three groups according to the types of spaces they explore, i.e., the spectral-based networks [10], [11], the spatial-based networks [12]–[14], and the spectral-spatial-based networks [15]–[18]. By either increasing the *representative power* [19]–[21] or *discriminative power* [5], [22], [23] of the features, these methods could achieve very promising state-of-the-art classification performance.

Nonetheless, deep learning-based classification suffers from deficiencies inherent with any supervised deep learning practice and HSI classification in particular, i.e., 1) a considerable amount of samples have to be labeled before training to prevent overfitting, and more importantly and 2) even if we take the time and effort to label a large amount of samples to train a classifier, such a classifier may perform poorly if the test images are from a *different* domain, i.e., images taken under different acquisition conditions. This is mainly because there exists large "spectral distribution variability"

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Ying Qu, Razieh Kaviani Baghbaderani, and Hairong Qi are with the Advanced Imaging and Collaborative Information Processing Group, Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN 37996 USA (e-mail: yqu3@vols.utk.edu; rkavian1@vols.utk.edu; hqi@utk.edu).

Wei Li and Yuxiang Zhang are with the School of Information and Electronics, Beijing Institute of Technology, Beijing 100081, China (e-mail: liwei089@ieee.org; zyx829625@163.com).

Lianru Gao is with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, Chinese Academy of Sciences, Beijing 100094, China (e-mail: gaolr@aircas.ac.cn).

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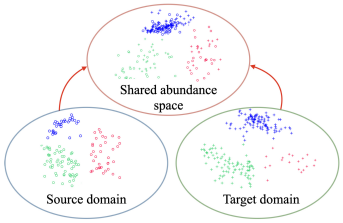


Fig. 1. Transfer learning of domain-invariant shared intrinsic representations through the shared abundance space.

among the same class of samples in different domains, due to changes in illumination, environmental, atmospheric, and temporal conditions [4], [5].

This, naturally, raises a question: *can we take advantage of the existing labeled data and acquire prior knowledge from them to classify images from a different domain?*

In the following, we define the domain of the existing labeled data as the *source domain* and the domain of the test data taken under different conditions as the *target domain*. There have been a few transfer learning [4], [24], [25] or active deep learning-based [5], [26], [27] methods specifically designed to address the above problem. They transfer the prior knowledge learned from the source domain to the target domain by *pretraining* in the source domain and *fine-tuning* in the target domain. These methods could achieve competitive performance and largely reduce the number of required labeled data in the target domain. However, the learning strategy based on fine-tuning is far from perfect. In order to fine-tune the network, a considerable amount of data in the target domain has to be labeled and nonnegligible computational resources for additional optimization steps are required to retrain the whole network [4], [5], [28].

This article aims to design a new transfer learning scheme to bridge the gap between the source and target domains without extra efforts required for data labeling and network retraining in the target domain. This is done by minimizing the domain discrepancy in a potential shared space, i.e., the abundance space, among different domains.

Due to the resolution issue, each pixel in an HSI usually covers a large geographical area, resulting in the so-called "mixed pixel", i.e., each pixel tends to cover more than one constituent material. These mixtures are generally assumed to be a linear combination of a few spectral bases with the corresponding coefficients (or abundance) satisfying two physical constraints, i.e., the sum-to-one constraint and the nonnegative constraint. For the pixels belonging to the same class in different domains, although they may possess different spectral characteristics, their abundance, indicating how much contribution each spectral basis has in constructing a certain pixel, should remain similar. Based on this hypothesis, we propose a physically constrained transfer learning method to address the challenges of HSI classification, referred to as physically constrained transfer learning through shared abundance space (PCTL-SAS).

The essential contribution of this work is the realization of the transfer learning scheme that can project the HSI data from the source and target domains into a shared abundance space based on their physical characteristics, as shown in Fig. 1. With this scheme, the network is able to extract domain invariant *shared intrinsic representations*, such that the classifier trained in the source domain can be directly utilized to categorize pixels in the target domain without additional efforts for data labeling or retraining. The novelty of this work is threefold.

1) First, to bridge the gap between the source and target domains, the proposed method projects the images from both domains into a shared abundance space with a shared Dirichlet-encoder that extracts representations following Dirichlet distribution which naturally meets the nonnegative and sum-to-one physical constraints. Based on the fact that each pixel usually consists of only a few bases, the abundance space is further constrained with the sparse entropy function to extract more effective representations.
2) Second, we propose an affine-transfer decoder that learns the potential physical affine correlations between the bases of different domains with a transfer function. The transfer function is further regularized by the proposed mutual discriminative network to match the correspondence between the pixels belonging to the same class in different domains, according to their uncertainties. In this way, the matched shared intrinsic representations can be extracted from different domains.
3) Third, to utilize the information from the different dimensional spaces of a given pixel, a densely connected 3-D-convolutional neural network (CNN) classifier is proposed and concatenated to the shared encoder. In this way, both the representative and discriminative power of the network is enhanced. What's more important is that the classifier is built on the domain-invariant shared abundance space, such that it can be trained with the labeled data in the source domain that still works in the target domain without additional data labeling or retraining.

The rest of this article is organized as follows. Section II provides an overview of the state-of-the-art HSI classification approaches. Section III provides the general formulation of the transfer learning problem and discusses the rationale of the proposed approach. Section IV elaborates the proposed PCTL-SAS method. Section V performs comprehensive evaluations of the proposed method. Conclusions are drawn in Section VI.

II. Related Work

A. HSI Classification

HSI classification is one of the most vibrant fields in remote sensing community and has been developed for the last four decades [5]. At first, traditional methods were mainly trained based on the pixel-wise strategy, which explores the spectral information of individual pixels. Numerous pixel-wise classifiers have been developed, among which, support vector machine (SVM) [29], [30], k-nearest-neighbor [31], maximum likelihood criterion [32], and logistic regression [33] are among the more popular. To improve classification accuracy, an instrumental way is to extract effective features or reduce the dimension of the HSI with different techniques before classification, e.g., principal component analysis (PCA) [34],

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[35], linear discriminative analysis (LDA) [36], independent component analysis (ICA) [37], and minimum noise fraction (MNF) [38], to name a few [39], [40]. Nonetheless, these methods rely heavily on the hand-crafted features designed for a specific task, which limited their ability to handle complex scenarios.

The performance of HSI classification has been boosted with the emergence of deep learning [4], [5]. The deep-learning-based approaches are able to extract hierarchical features automatically in an end-to-end manner, which demonstrated a unique strength in HSI classification. Generally, existing deep-learning-based approaches can be categorized into three groups according to the types of HSI spaces they explore, i.e., the spectral-based methods, the spatial-based methods, and the spectral–spatial-based methods [5]. Spectral-based methods were developed based on 1-D CNN architecture to perform pixel-wise classification according to the spectral characteristics of HSI [10], [11]. Although 1-D-CNN-based methods could achieve better results compared to those of the traditional methods, the correlation information among spatially adjacent pixels were not fully explored. Spatial-based methods further improved the classification accuracy by applying 2-D-CNN on the patch surrounding the given pixel to utilize its spatial adjacency information [12]–[14]. In order to fully utilize the spectral–spatial correlation information, spectral–spatial-based methods were developed by adopting the 3-D convolutional kernels and demonstrated promising classification results in recent studies [15]–[18], [41], [42]. For example, Spec-Spat [41], 3-D-CNN [16], and SSRN [15] were designed with different network structures but all based on 3-D-CNN blocks. HSI-CNN [17] and HybridSN [18] were constructed with both the 3-D-CNN and 2-D-CNN basic blocks.

All these state-of-the-art approaches could achieve promising results when the training and testing are performed in the same domain. However, they may perform poorly on a different domain due to spectral variability, even if the labeled classes are shared among different domains.

B. HSI Transfer Learning

Recently, a few transfer-learning-based methods were proposed attempting to address the challenges mentioned above. Generally, given a sufficient amount of labeled data from both the source and target domains, these methods solve the HSI classification problem by adopting a transfer learning strategy, which trains the model on the source domain and fine-tunes such a model on the target domain. Liu *et al.* [43] constructed multiple geodesic flows using pairs of spatial and spectral features obtained from both the source and target domains that were further fed to an SVM to complete the classification task. Zhang *et al.* [25] utilized a 3-D lightweight CNN structure and performed two transfer learning strategies, including cross-sensor and cross-modal strategies. HT-CNN [44] reused the priors learned with low-level and mid-level of VGGNet [45] and adopted an attention mechanism to adjust the feature maps due to the difference between the heterogeneous data sets.

Another set of works employ the active learning strategy to manually annotate a small number of samples in the target domain to boost the classification performance. Deng *et al.* [24] sequentially retrained a multikernel classifier with a collection of the available data from the source domain and the added, most informative samples acquired with active queries from the target domain. The work of [27] extracted the salient samples from both the source and target domains, explored the correlation of these domains through a deep network layer-by-layer analysis, and further fine-tuned the network. Deng *et al.* [26] proposed a hierarchical stacked sparse autoencoder to learn joint spectral–spatial features which are later transferred to the target domain by fine-tuning the network using a few labeled samples from both the source and target domains.

Although these methods could classify images in the target domain successfully, the learning strategy based on fine-tuning is far from perfect. In order to fine-tune the models, a sufficient amount of data samples from both the source and target domains have to be labeled and the whole network has to be retrained which requires a nonnegligible computational resource [4], [5].

III. Problem Formulation and Motivations

Given n samples, $X_S = \{\mathbf{x}_S | \mathbf{x}_{Si} \in \mathcal{X}_S, i = 1, \ldots, n_S\}$, in the source domain, $\mathcal{X}_S$, and the corresponding label set $Y_S = \{y_S | y_{Si} \in \mathcal{Y}_S, i = 1, \ldots, k_S\}$ in the source label domain, $\mathcal{Y}_S$, we could build a training set and learn a classifier $f_S : \mathcal{X}_S \rightarrow \mathcal{Y}_S$. However, the classifier f_S trained in the source domain may fail to categorize pixels, $X_T = \{\mathbf{x}_T | \mathbf{x}_{Ti} \in \mathcal{X}_T, i = 1, \ldots, n_T\}$, in the target domain, $\mathcal{X}_T$, due to the distribution discrepancy, i.e., $p(X_S) \neq p(X_T)$. In order to address this problem, previous work usually retrains the whole network in the target domain $\mathcal{X}_T$ with weights initialized by f_S in the source domain [4], [5]. However, to retrain the network in the target domain, $\mathcal{X}_T$, we have to label a sufficient amount of data in $\mathcal{X}_T$, which would introduce additional nontrivial and time-consuming burden.

In this article, we seek to bridge the gap between the source and target domains by realizing a new transfer learning scheme through a shared space. By projecting the data from both domains into a proper shared space $\mathcal{A}$, we aim to minimize the domain discrepancy and train a general classifier f_{ST} that works in both domains. Hence the first challenge is how to find such a shared space given only the labels in the source domain.

We approach this problem from a unique angle, i.e., by exploiting the physical characteristics inherent in data in both the source and target domains. As discussed in Section I, each pixel in an HSI tends to cover more than one constituent materials. Thus, these mixtures can be modeled as a linear combination of a few spectral bases with the corresponding mixing coefficients (or abundance). Mathematically, they can be expressed as

$$\mathbf{x}_S = \mathbf{a}_S B_S \tag{1}$$

$$\mathbf{x}_T = \mathbf{a}_T B_T \tag{2}$$

where $\mathbf{a}_S \in \mathbb{R}^{1\times c}$ and $B_S \in \mathbb{R}^{c\times l}$ denote the coefficient vector (or abundance vector) and the spectral bases of the given pixel $\mathbf{x}_S \in \mathbb{R}^{1\times l}$ in the source domain, respectively.

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$\mathbf{a}_T \in \mathbb{R}^{1\times c}$ and $B_T \in \mathbb{R}^{c\times l}$ denote the abundance vector and the spectral bases of the given pixel $\mathbf{x}_T \in \mathbb{R}^{1\times l}$ in the target domain, respectively. l and c denote the number of spectral bands of the HSI and the number of the spectral bases, respectively. Assume that $\mathbf{x}_S$ and $\mathbf{x}_T$ belong to the same class, and they may behave distinctively due to different illumination conditions, and so do their spectral bases $\mathbf{a}_S$ and $\mathbf{a}_T$. However, for the same class, the abundance vectors $\mathbf{a}_S$ and $\mathbf{a}_T$, indicating the proportions of the bases, should be similar in both domains, hence serving as the potential shared representations and forming the corresponding shared space to facilitate the training of a general classifier f_{ST}. We refer to this space as the shared abundance space. See Section V-D for an illustration of the effectiveness of the abundance space representation. Given such a potential shared space, the second challenge is then how to project the data from both domains into the shared abundance space, such that the shared representations could be extracted.

We tackle this problem with three physical properties inferred from the characteristics of the data, i.e., 1) the nonnegative and sum-to-one constraints on the abundance $\mathbf{a}_S$ and $\mathbf{a}_T$; 2) the affine correlations between B_S and B_T; and 3) the uncertainty residing in the usage of $\mathbf{a}_S$ ($\mathbf{a}_T$) to represent $\mathbf{x}_S$ ($\mathbf{x}_T$). Since there is no ground truth label in the target domain, we need to exploit these intrinsic properties of the data in order to regulate the solution space. The first item is inferred from the properties of the abundance. Since the abundance vector indicates the proportions the spectral bases in making up the mixed pixels, the shared space should be constrained with the nonnegative and sum-to-one physical constraints. The second item is inferred from the properties of the spectral bases. Although the spectral bases of the mixed pixels belonging to the same classes may be distinctive, they tend to hold an affine relationship [46], that is,

$$B_S = \mathbf{c}B_T + \mathbf{d}. \tag{3}$$

In addition, since the shared abundance vector can represent both $\mathbf{x}_S$ and $\mathbf{x}_T$, it has the potential to be the optimal representation that maximizes the reduction of the uncertainty of both $\mathbf{x}_S$ and $\mathbf{x}_T$.

IV. Proposed Approach

We propose a physically constrained transfer learning approach for HSI classification across different domains through the extraction of the shared abundance space, as shown in Fig. 2. The network architecture mainly consists of four unique modules, i.e., 1) a shared sparse Dirichlet-encoder E_ϕ which projects the data from different domains into a shared abundance space with the nonnegative and sum-to-one physical constraints; 2) an affine-transfer decoder D_ψ to embed the potential affine relationship between the source and target domains; 3) a mutual discriminative network I_ω to enforce the correspondence between the pixels in the source and target domains such that the shared intrinsic representations could be extracted from pixels belonging to the same class; and 4) a densely connected 3-D-CNN-based classifier C_σ concatenated with the shared encoder to increase the generalization, representative, and discriminative power of the network.

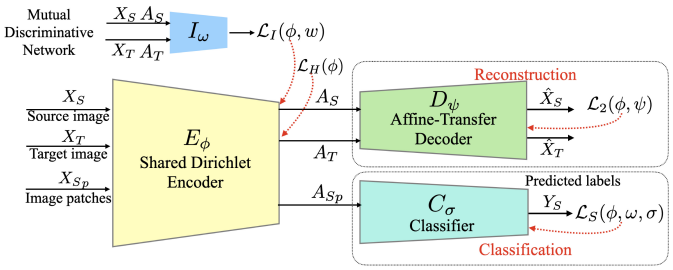


Fig. 2. Flowchart of the proposed physically constrained transfer learning approach for HSI classification.

A. Architecture

To realize the transfer learning scheme for effective generalization, both the representation and discrimination power of the network need to be boosted.

The effective representation is realized with the proposed encoder-decoder structure consisting of E_ϕ and D_ψ. This structure is designed according to the linear mixing model shown in (1) and (2), where the representations extracted by the encoder E_ϕ correspond to the abundance of the images and the decoder D_ψ embeds the spectral bases of the images in (1) and (2). Let us define the input image domain of the encoder-decoder network as $\mathcal{G} = \{X_S, X_T\}$, where $X_S \in \mathcal{X}_S$, $X_T \in \mathcal{X}_T$ denote the source image and the target image, respectively. Similarly, the output domain is denoted with $\hat{\mathcal{G}} = \{\hat{X}_S, \hat{X}_T\}$, where $\hat{X}_S$ and $\hat{X}_T$ denote the reconstructed source and target images, respectively. The shared Dirichlet encoder $E_\phi : \mathcal{G} \to \mathcal{A}$, projects the input data from the image domain to the so-called "abundance space" $\mathcal{A} = \{A_S, A_T\}$, where the representations A_S and A_T are extracted (latent variables on the hidden layer) from the source and target images, respectively. The affine-transfer decoder $D_\psi : \mathcal{A} \to \hat{\mathcal{G}}$ reconstructs the images back from the representations to the output image domain $\hat{\mathcal{G}}$ with the spectral bases of both the source and the target images.

The affine-transfer decoder of the network not only learns the spectral bases of the images in the source and target domains, but also finds their potential affine relationship as demonstrated in (3). To further regularize the representations influenced by the affine relationship, we propose a mutual discriminative network I_ω and concatenate it with the encoder. I_ω maximizes the mutual information (MI) between the representations A_S (A_T), and their own corresponding input X_S (X_T). In this way, the shared intrinsic representations are able to be extracted between the pixels belonging to the same classes in different domains.

Besides the effective representation scheme described above, the network architecture also realizes effective discrimination by the proposed densely connected 3-D CNN, C_σ, which is concatenated to the shared Dirichlet-encoder. The classifier is trained with the patches of the source images X_{S_p} to utilize both the spatial and spectral information of the data. It can be defined as $f_S : X_{S_p} \to A_{S_p} \to \mathcal{Y}_S$,

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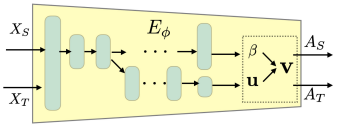


Fig. 3. Structure of the proposed shared sparse Dirichlet encoder.

which is built as a mapping function between the inputs of the source image X_{S_p} and their labels Y_S. By sharing the same encoder, the classifier C_σ further increases the discriminative power of the network. More important, since the weights of the classifier are updated on the domain-invariant abundance space $\mathcal{A}$, the generalization capacity of the classifier is enhanced so that it can be directly utilized to categorize target images without data labeling or network retraining.

The detailed network structure is further elaborated in Sections IV-B–IV-D. Section IV-E also includes implementation details.

B. Shared Sparse Dirichlet Encoder

As discussed in Sections I and III, both the source image X_S and the target image X_T can be represented by a linear combination of the spectral bases and their corresponding abundance vectors as shown in (1) and (2). Although the spectral bases may vary due to spectral variability [46], their abundance vectors, which indicate how much each spectral base contributes to a given pixel should be the same for the same class. Thus, the abundance vectors indicate the domain-invariant representations we aim to find. Refer to Section V-D for an illustration of the effectiveness of abundance-based representation. Due to the physical properties of the abundance, the representations should meet two physical constraints, i.e., nonnegativity and sum-to-one. We extract such representations by projecting the data from both the source and target domains to the shared abundance space, with sparse Dirichlet-encoder as shown in Fig. 3.

The Dirichlet-encoder is constructed with the stick-breaking process [47], [48], which can be illustrated as breaking a unit-length stick into c pieces. The pieces would follow a Dirichlet distribution, and thus the two physical constraints, nonnegativity and sum-to-one are naturally met. A single piece, a_j, can be expressed as

$$a_j = \begin{cases} v_1, & \text{for } j = 1 \\ v_j \prod_{o<j}(1 - v_o), & \text{for } j > 1 \end{cases} \tag{4}$$

where v_j is drawn from the inverse transform of the Kumaraswamy distribution [49], which is defined as

$$\text{kuma}(u, \alpha, \beta) = \alpha\beta u^{\alpha-1}(1 - u^\alpha)^{\beta-1}. \tag{5}$$

The Kumaraswamy distribution is equivalent to Beta distribution when $\alpha = 1$ or $\beta = 1$. Let $\alpha = 1$, v_j can be drawn from its inverse transformation defined in (6)

$$v_j \sim \left(1 - (1 - u_j)^{\frac{1}{\beta_j}}\right). \tag{6}$$

The stick-breaking process in (4) can be explained as sequentially breaking v_j from the remaining stick $\prod_{o<j}(1 - v_o)$. The constructed representation, $0 \le a_j \le 1$, would naturally meet the nonnegativity constraint. As the number of pieces c increases, the summation of a_j is close to one, i.e., $\sum_{j=1}^{j=c} a_j \approx 1$, which would enforce the representations to meet the sum-to-one physical constraint.

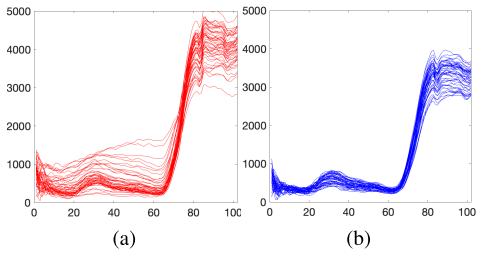


Fig. 4. Spectral reflections of data from the same class in (a) source domain and (b) target domain.

Based on the fact that each pixel usually consists of a few spectral bases, we further encourage the representations to be sparse with the normalized entropy function [50], defined in (7) with $p = 1$

$$H_p(\mathbf{a}) = -\sum_{i=1}^{N} \frac{|a_{i\rightarrow}|^p}{\|a_{i\rightarrow}\|_p^p} \log \frac{|a_{i\rightarrow}|^p}{\|a_{i\rightarrow}\|_p^p}. \tag{7}$$

Even if a vector is sum-to-one, its normalized entropy function could decrease monotonically when it becomes sparse. This could not be satisfied with traditional l_1 regularization or Kullback–Leibler (KL) divergence, because they enforce the sparsity by reducing the summation of representations which would be approximated to 1 all the time due to the stick-breaking structure.

The objective function for sparse loss can then be defined as

$$\mathcal{L}_H(\phi) = H_1(E_\phi(X_S)) + H_1(E_\phi(X_T)). \tag{8}$$

C. Affine-Transfer Decoder and Mutual Discriminative Network

With the shared sparse Dirichlet-encoder, we are able to project the data from the image domain to the abundance space. However, since data distributions of the source and target domains are different and there is no label information for the images in the target domain, we still need to match data belonging to the same classes but in different domains to extract their effective shared intrinsic representations. For this purpose, more physical constraints are exploited to further regularize the solution space. These include the affine correlations between different domains and the uncertainty reduction of the representations. In our network design, we enforce these two physical constraints with the proposed affine-transfer decoder and the mutual discriminative network.

1) Affine-Transfer Decoder: Due to different environmental and illumination conditions during data acquisition, the HSI pixels belonging to the same class but different domains may possess different spectral characteristics as shown in Fig. 4. We can observe that although the reflectances in the source and

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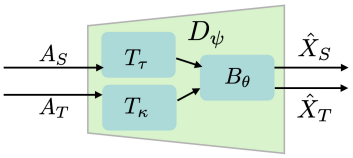


Fig. 5. Structure of the proposed affine-transfer decoder.

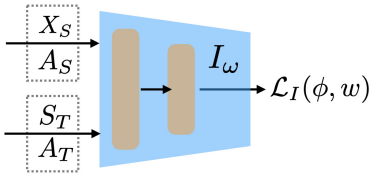


Fig. 6. Structure of the proposed mutual discriminative network.

target domains are different, their relationship can be approximated with an affine transfer, i.e., the spectral reflectance in one domain is able to be represented as the affine transformation of that in another domain. Similarly, their spectral bases can be modeled with an affine relationship as shown in (3). Since the pixels belonging to the same classes in different domains have different distributions, if we use a general decoder, the difference between the distributions may be reflected on the representations, making it difficult to find the shared representations. In this article, we propose an affine-transfer decoder to address this problem. The affine-transfer decoder is built to learn the spectral bases of both domains and simulate their physical affine relationship. In this way, the network has the potential to learn the shared intrinsic representations. The network structure is shown in Fig. 5.

To improve the flexibility of the network, instead of directly adopting the affine-transfer model, we introduce a shared basis B_θ to build the relationship between the spectral bases B_S and B_T of the source and target domains. Then the affine-transfer decoder is defined as

$$T_\tau(B_\theta) = \mathbf{c}_\tau B_\theta + \mathbf{d}_\tau \tag{9}$$

$$T_\kappa(B_\theta) = \mathbf{c}_\kappa B_\theta + \mathbf{d}_\kappa \tag{10}$$

where $\mathbf{c}_\tau$, $\mathbf{d}_\tau$ and $\mathbf{c}_\kappa$, $\mathbf{d}_\kappa$ are the network weights. $T_\tau(B_\theta)$ and $T_\kappa(B_\theta)$ share the same basis weights B_θ and they correspond to B_S and B_T in (1) and (2), respectively. The affine-transfer decoder can not only extract the spectral bases from images on both domains but also learn their potential affine relationship. Compared to the encoder–decoder network with general decoder, the proposed model with the affine-transfer decoder allows the network to extract shared intrinsic representations even if they hold different statistic distributions.

2) Mutual Discriminative Network: With the affine-transfer decoder, the network has the potential ability to extract shared intrinsic representations. However, to avoid negative transfer, the affine-transfer relationship should be built between the same classes in different domains. We enforce such correspondence by regularizing the solution space by maximizing the MI between the representations and their raw input. This is realized by the proposed mutual discriminative network. The network structure is shown in Fig. 6.

In our network design, the classification accuracy would rely heavily on the extracted representations in the abundance space. The optimal representations should not only reconstruct the image well but also reduce the uncertainty of the image to the maximum extent. By finding the optimal representations for each pixel, the shared representations could be automatically matched between the pixels belonging to the same class in different domains. We introduce MI to maximize the reduced uncertainty of the image. MI has been widely used for multimodality registrations. It measures how much uncertainty of one variable is reduced given the other variable with Shannon entropy. The shared intrinsic representation should reduce the uncertainty of both the source and target domains. Based on the work of Belghazi *et al*. [51], the MI can be estimated through a neural network according to the lower bound of KL-divergence based on the Donsker-Varadhan (DV) representation. Thus, we also define a discriminative network I_w to maximize the average MI between each representation and its own input, i.e., $I_\omega(X_S, E_\phi(X_S))$ and $I_\omega(X_T, E_\phi(X_T))$. However, in our case, instead of estimating the exact MI, we only use MI to regularize the solution space. Therefore, an alternative lower bound based on Jensen-Shannon is introduced, which is more stable than the DV-based objective function [52].

Taking the image X_S in the source domain as an example, the objective function for the mutual discriminative network $I_\omega : \mathcal{G} \times \mathcal{A} \to \mathbb{R}$ is defined as

$$\begin{aligned}\mathcal{I}_{\phi,\omega}(X_S, E_\phi(X_S)) = E_{\mathbb{P}}[-sp(-I_\omega(X_S, E_\phi(X_S))] \\ - E_{\mathbb{P}\times\tilde{\mathbb{P}}}\left[sp\left(I_\omega\left(X'_S, E_\phi(X_S)\right)\right)\right]\end{aligned} \tag{11}$$

where ω denotes the network weights, and $sp(x) = \log(1 + e^x)$. Note that X'_S is an input with samples drawn from the distribution $\tilde{\mathbb{P}} = \mathbb{P}$, which is a negative sample generated by randomly shuffling the input data. The objective MI function for both the source and target image is defined as

$$\mathcal{L}_I(\phi, \omega) = \mathcal{I}_{\phi,\omega}(X_S, E_\phi(X_S)) + \mathcal{I}_{\phi,\omega}(X_T, E_\phi(X_T)) \tag{12}$$

D. Densely Connected 3-D-CNN-Based Classifier

To train a classifier that works in both domains with better classification accuracy, a densely connected 3-D CNN-based classifier C_σ is proposed and concatenated to the shared Dirichlet-encoder. In this way, the classifier is mainly trained on the domain invariant abundance space, which would largely increase the generalization of the network. The network structure of the classifier is shown in Fig. 7.

The basic block of the classifier consists of a 3-D-CNN, a batchnorm layer [53] and a relu activation function [54] defined in (13), as shown in the lower part of Fig. 7. With 3-D-CNN, the network is able to categorize a pixel not only considering its own information but also its adjacent pixels. The batchnorm and the relu are used to accelerate the training speed. To increase the discriminative power of the network, the 3-D-CNN blocks are densely connected, i.e., each layer is fully connected with all its subsequent layers

$$\text{relu}(y) = \max(0, y). \tag{13}$$

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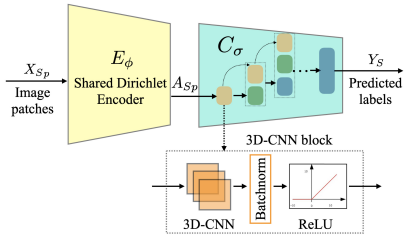


Fig. 7. Structure of the proposed densely connected 3-D-CNN classifier.

Algorithm 1 PCTL-SAS

Input: source data set $X_S \in \mathcal{R}^{m\times n\times l}$, target data set $X_T \in \mathcal{R}^{M\times N\times l}$, n_S patch samples from the source data set $X_{Sp} = \{\mathbf{x}_{Sp_i}|\mathbf{x}_{Sp_i} \in \mathcal{R}^{7\times 7\times l}, i = 1,\ldots,n_S\}$, and their corresponding label set $Y_S = \{y_{S_i}|i = 1,\ldots,n_S\}$

Output: Predicted labels for the target data set $Y_T = \{y_{T_i}|i = 1,\ldots,n_T\}$

Begin:

1) Training:

1) Set up the designed model PCTL-SAS with both the reconstruction network constructed with fully connected layers and the classification network built with 3-D-CNN blocks, as shown in Fig. 2.
2) Feed in the X_S, X_T,X_{Sp} into the network and perform backpropagation with the objective function defined in (17), with reconstruction loss, sparse constraints, mutual discriminative loss, and classification loss.

2) Testing: Feed in the n_T patch samples from the target data set $X_{Tp} = \{\mathbf{x}_{Tp_i}|\mathbf{x}_{Tp_i} \in \mathcal{R}^{7\times 7\times l}, i = 1,\ldots,n_T\}$ and predict their labels Y_T with the proposed shared Dirichlet encoder and classifier.

End

The last layer of the classifier is a fully connected layer following a dropout operation [55] to prevent overfitting. A softmax activation function defined in (14) and cross entropy objective function are introduced to predict the class label

$$\text{softmax}(y_j) = \frac{e^{y_j}}{\sum_{j=1}^{\text{num of classes}} e^{y_j}}. \tag{14}$$

Given the ground truth label, Y_S, from the source domain and the predicted label $Y_{PS} = C_\sigma(E_\phi(X_{Sp}))$ from the network, the cross-entropy objective function is defined as

$$\mathcal{L}_S(\phi,\sigma) = -\sum Y_S \log(Y_{PS}). \tag{15}$$

E. Implementation Details

The objective function of the proposed architecture includes four components, a reconstruction loss $\mathcal{L}_2(\phi,\psi)$, a sparse constraint $\mathcal{L}_H(\phi)$, the mutual discriminative loss $\mathcal{L}_I(\phi,\omega)$, and a classification loss $\mathcal{L}_S(\phi,\sigma)$. The reconstruction loss is defined as

$$\mathcal{L}_2(\phi,\psi) = \|D_\psi(E_\phi(X_S)) - X_S\|_2 + \|D_\psi(E_\phi(X_T)) - X_T\|_2 \tag{16}$$

where $\|\cdot\|$ denotes the l_2 norm. The definition of the other losses are defined in (8), (12), and (15), respectively. The objective function for the proposed network can then be expressed as

$$\mathcal{L}(\phi,\psi,\omega,\sigma) = \mathcal{L}_2(\phi,\psi) + \alpha\mathcal{L}_H(\phi) + \lambda\mathcal{L}_I(\phi,\omega) + \mathcal{L}_S(\phi,\sigma) \tag{17}$$

where α and λ are the parameters that balance the trade-off among different losses. Since the reconstruction and classification loss are equally important, their parameters are fixed to 1.

The proposed method is an end-to-end network as shown in Fig. 2, which consists of two branches, i.e., 1) a reconstruction branch and 2) a classification branch. These two branches share the same Dirichlet encoder, i.e., both the affine-transfer decoder and the classifier are concatenated with the shared encoder. The network is optimized by back-propagation given the objective function defined in (17). The path of the back-propagation is illustrated with four red dashed lines in Fig. 2. The top three dashed lines illustrate the paths for the reconstruction loss, the sparse loss, and the mutual discriminative loss, respectively. The bottom dashed line indicates the path for the classification loss. By optimizing the entire network, it not only enhances the representative power of the network but also the discriminative power of the network. Note that, the inputs for the reconstruction branch are the source and target images, while the input to the classification branch is the patches of the source images. The whole procedure for training and testing are summarized in Algorithm 1.

During the training procedure, we are able to extract the shared intrinsic representations and train a general classifier on them. In the testing procedure, we can directly get the predicted labels by feeding the target image patches into the network. That is, no data labeling or network retraining is required to classify the image in the target domain.

The reconstruction branch is constructed with a few fully connected layers, and the classifier consists of a few 3-D-CNN layers with kernel size $3\times 3\times 3$. The number of layers and nodes in the proposed network are shown in Table I. The encoder of the network has six layers with two branches for u and β. The first three layers are shared by u and β as shown in Fig. 3. Except for the last layer, the number of nodes for layers is increased from 3 to $3\times \ln$, where ln denotes the layer number. The last layer before u and β contain 11 nodes and 1 node, respectively. The affine-transfer decoder is constructed with shared basis $\mathcal{B}_\theta$ and two transfer layers, i.e., T_τ and T_κ. $\mathcal{B}_\theta$ has two layers, and each layer has 11 nodes. Both T_τ and T_κ have one layer with 11 nodes. The mutual discriminative network consists of two layers with 13 nodes and 1 node, respectively. The classifier is constructed with 5 layers, and the number of nodes in each layer is shown in Table I. The basic block of each layer consists of a 3-D-CNN layer, a batchnorm layer, and a relu activation function as shown in the lower part of Fig. 7.

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TABLE I
NUMBER OF LAYERS AND NODES IN THE PROPOSED NETWORK

	E_ϕ	I_w	T_τ/T_κ	B_θ	3D-CNN
#layers	6	2	1/1	2	5
#nodes	$3\times ln$, $ln=\{1,2,3,4,5,6\}$	[13,1]	[11]	[11]	[12,32, 12,12,30]

TABLE II
DATA SET DESCRIPTION FOR THE SOURCE IMAGE, PU AND THE TARGET IMAGE, PC

No.	Class	Source: PU	Target: PC
1	Trees	3064	7598
2	Bare soil	5029	6584
3	Bitumen	1330	7287
4	Bricks	3682	2685
	Total	13105	24154
	1%	131	0
	3%	393	0
	5%	655	0

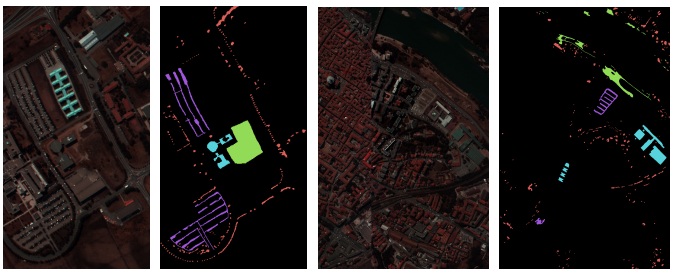
V. EXPERIMENTS AND RESULTS

A. Data Sets

To evaluate the performance of the proposed PCTL-SAS, we apply the method on three sets of source and target image pairs with different spectral characteristics that bring different levels of challenges to the classification problem. For each data set pair, we choose the image with a smaller number of samples as the source image and the other one as the target image. The common classes shared by both the source and target images are selected for performance evaluation. To visualize the different levels of challenges brought by the three sets of data, we project the three data set pairs into a 2-D space using the singular value decomposition (SVD) method, as demonstrated in Figs. 16(a), 17(a), and 18(a), respectively. We can observe that there are some overlaps between the spectra of the same class on the source and target domains on the first data set, while on the second and third data set pairs, there exists a large discrepancy between the same class on the source and target domains. This domain discrepancy indicates that the second and third data sets are more challenging than the first one. The details of the data set pairs are described below.

1) PaviaU-PaviaC: The PaviaU-PaviaC data pair consists of the source image, Pavia University (PU), and the target image, Pavia Center (PC). The PU and PC images were acquired by the Reflective Optics System Imaging Spectrometer (ROSIS) sensor over Pavia, Italy [56]. The spatial resolution of both images is 1.3 m. The PU image has the dimension of 610×340, with 102 bands, and the last band removed. The PC image has the dimension of 1096×715 with 102 bands. Four shared classes, tree, bare soil, bitumen, and bricks are chosen among the source and target images to evaluate the proposed method. The number of samples corresponding to each class is listed in Table II and a sample pair of images are shown in Fig. 8.

2) Houston: The Houston pair is composed of the source image, Houston-2013 (H13) [57], and the target image Houston-2018 (H18) [58], which were collected over the



(a) (b) (c) (d)

Fig. 8. (a) Source image PU. (b) Ground truth of PU. (c) Target image PC. (d) Ground truth of PC.

TABLE III
DATA SET DESCRIPTION FOR THE SOURCE IMAGE, HOUSTON-2013 (H13), AND THE TARGET IMAGE, HOUSTON-2018 (H18)

No.	Class	Source: H13	Target: H18
1	Grass healthy	345	2099
2	Grass stressed	375	7693
3	Trees	365	3128
4	Water	285	30
5	Residential buildings	313	8271
6	Non-residential buildings	408	49093
7	Road	439	9558
	Total	2530	79872
	1%	25	0
	3%	75	0
	5%	126	0

TABLE IV
DATA SET DESCRIPTION FOR THE SOURCE IMAGE, SHANGHAI, AND THE TARGET IMAGE, HANGHZOU

No.	Class	Source: Hanghzou	Target: Shanghai
1	Water	18043	123123
2	Ground/Building	77450	161689
3	Plants	40207	83188
	Total	135700	368000
	1%	1357	0
	3%	4071	0
	5%	6785	0

University of Houston campus at different times with spatial resolution of 2.5 m. The H13 and H18 have 144 and 48 bands, respectively. In H13, the overlapped 48 bands corresponding to that of the H18 are kept. The overlap area of two images are resized to the same dimension, 209×955. Seven shared classes are chosen. The images and the number of samples are shown in Fig. 9 and Table III, respectively.

3) Hanghzou–Shanghai: This pair of data sets, consisting of the source image, Hangzhou, and the target image, Shanghai, was acquired by the REO-1 Hyperion hyperspectral sensor [59]. The number of spectral bands obtained by the sensor is 220, of which 198 spectral bands are kept after removing 22 bad bands. The Hangzhou data set, with a dimension of 590×230, was acquired on 2 November 2002, over Hangzhou city. It covers the water areas of the West Lake and Qiantang River, buildings and plants. The Shanghai data set, with a dimension of 1600×230, was acquired on 1 April 2002, over Shanghai city, which is 170 km far away from Hangzhou. It covers buildings, plants, and the water areas of the Yangtze

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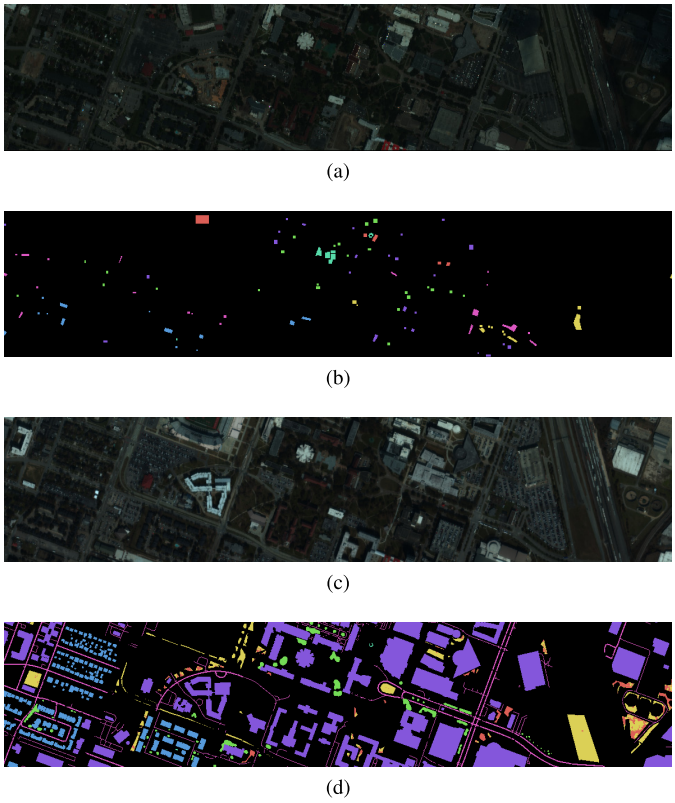


Fig. 9. (a) Source image Houston-2013 (HS13). (b) Ground truth of HS13. (c) Target image Houston-2018 (HS18). (d) Ground truth of HS18.

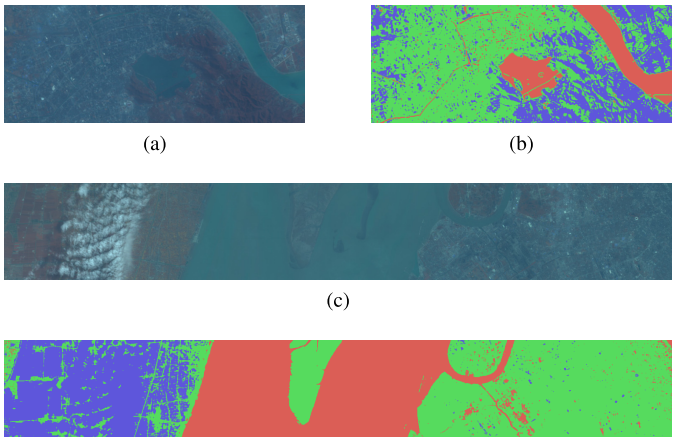


Fig. 10. (a) Source image Hangzhou. (b) Ground truth of Hangzhou. (c) Target image Shanghai. (d) Ground truth of Shanghai.

River and Huangpu River. The detail of the data sets can be found in Table IV and the images are shown in Fig. 10. Note that compared to the other two data sets, this data set is very noisy, hence it can be used to evaluate the robustness of different classifiers.

B. Experimental Setup

The classification results of the proposed approach are compared with eight state-of-the-art approaches including 1-D-CNN [11] based on 1-D CNN; Spec-Spat [41], 3-D-CNN [16], and SSRN [15] based on 3-D CNN; HSI-CNN [17] and HybridSN [18] based on both 2-D and 3-D CNN; and the transfer learning framework HT-CNN [44] based on VGGNet [45] and the cross-temporal method proposed by Liu *et al*. [43]. Note that since the code for Liu *et al*. [43] is not available, Note that since the code for Liu *et al*. [43] is not available, we simply list their result in overall classification accuracy on the shared Houston data set at the end of Section V-C.

All these methods are reported with the best performance in recent literature with available codes. To test the robustness of the methods, 1%, 3%, and 5% of the samples in the source data set are randomly selected as training samples. The trained classifiers are further applied to predict the labels of both the source and target data sets. For quantitative comparison, the classification performance is evaluated with three widely used metrics, i.e., overall accuracy (OA), average accuracy (AA), and kappa coefficient (κ) [4]. OA is estimated by dividing the number of corrected predicted labels by the number of total labels for the entire image. AA is the average classification accuracy for all the classes. κ is a statistical coefficient, which measures the degree of accuracy and reliability for classification. The higher the metrics, the better the classification performance.

We conduct two groups of classification experiments. In Section V-C, we thoroughly study the classification accuracy of the proposed scheme and compare to the seven state-of-the-art approaches. In Section V-D, we provide a visual inspection of the same class sample distributions in both the raw image space and the abundance space to qualitatively show the effectiveness of the proposed scheme. In Section V-E, we perform ablation study on the different components forming the proposed scheme.

C. Classification Results

We conduct two groups of experiments. The first group performs training and testing on the same data set to set up baseline. The second group performs training on the source data set and testing on the target data set.

1) Training and Testing on the Same Data Set: We first perform experiments on the general case where both the training and testing are conducted on the source data set. The methods are trained with 1%, 3%, and 5% of the labeled samples from each source data set. The classification results of different methods on the three data sets are reported in Tables V–VII, respectively. Since the spectral characteristics are diverse across different data sets, the performance of the same method may work well for one data set but fail on the other data set. Thus, to demonstrate the trends of the methods in different scenarios, we average each metric over the three data sets and show the results in Fig. 11.

We can observe that when the number of training samples increases, most methods achieve a better classification accuracy. The Spec-Spat [41], HSI-CNN [17] and especially the 3-D-CNN [16] methods are more sensitive to the number of training samples. They could achieve a relatively good accuracy when there is a sufficient number of training samples,

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e.g., on the first data set, PU, and the third data set, Hangzhou, as shown in Tables V and VII, respectively. However, their classification performance drops significantly on the second data set, Houston-2013, which has a limited number of training samples. We can observe that the OA of these methods is below 60% when we train the network with 1% of the labeled samples in the Houston data set. The 1-D-CNN [11] performs well with a limited number of training samples. However, since 1-D-CNN depends only on the spectral information of the pixels, its performance can be unstable due to noise. For example, in the third data set which is more noisy than the other two, when the number of training samples increases, the classification performance actually drops. The SSRN [15] and HybridSN [18] work well for the first two data sets as shown in Tables V and VI. However, they are not robust enough to noisy data. As shown from the results of the noisy data in Table VII, the performance of these two methods drops even the third data set which has more number of training samples than that of the first two data sets.

The transfer learning-based method HT-CNN [44] achieves better accuracy than that of Spec-Spat, HSI-CNN and 3-D-CNN. That is because the network is designed based on VGGNet which carries the image priors. However, the small number of training samples limited the classification performance of the HT-CNN due to its large number of network weights. The proposed PCTL-SAS achieves better or comparable accuracy to the other methods. As shown in Fig. 11, the average OA, AA and Kappa of the proposed method outperform the other methods to a large margin because of its strong representative and discriminative power. Note that, since the networks are trained to extract representations from both the source and the target domains, it sacrifices the accuracy on the source image a little bit. If we only need to classify the images in the source domain, we could remove the reconstruction branch for the target domain. In this way, the classification accuracy would be further increased.

2) Training on the Source Data Set and Testing on the Target Data Set: In the second group of experiments, the pre-trained models based on 1%, 3%, and 5% of the source labeled samples are adopted to classify the pixels of the target data sets without additional training. The classification results of different methods on the three target data sets are shown in Tables VIII–X, respectively. Similarly, the average metrics over the three data sets are illustrated in Fig. 12. The best classification map of each method on the three target data sets are shown in Figs. 13–15, respectively.

We can observe that the accuracy of all the methods decreases in this challenging task due to the pixel variability caused by different acquisition conditions. The 1-D-CNN method works well on the first target data set, but its performance drops on the second target data set and fails on the third target data set as shown in Fig. 15. Although SSRN [15] and HybridSN [18] work better than most of the methods on the first source data set, they could not achieve comparable results to that of the Spec-Spat, HSI-CNN, and 3-D-CNN methods on the first target data set as shown in Fig. 13. This is because there exists spectral variability between the same classes in different domains, and the methods could not model the intrinsic representations among different domains. The same can be observed on the second and third data sets, as shown in Figs. 14 and 15.

It is worth mentioning that since HybridSN projects the data set from the image domain to the low-dimensional domain with PCA before classification, it removes some noise and achieves better accuracy than most of the methods on the third target data set. However, the PCA projection is not suitable for the first and second data sets, because the information shared between the source and target data sets tends to be missed after applying PCA on the data sets with low noise. Thus, it fails on these two data sets as shown in Figs. 13 and 14. The HT-CNN methods could achieve better results than some of the other methods due to its transformed image priors, but it could not maximize its efficiency without involving training samples on the target domain. The proposed method projects data from the image domain to the shared abundance space with the affine-transfer decoder and mutual discriminative network, thus it achieves promising results and outperforms the other methods by a large margin on the target data sets as shown in Fig. 12.

Note that we also compare the proposed method with the cross-temporal method proposed by Liu *et al*. [43]. Since the code for Liu *et al*. [43] is not available, we directly list their result on the shared Houston data set for a fair comparison. With the model trained with 100% training samples on the source data set, Houston-2013, Liu *et al*. [43] achieves the OA of 67% on the target data set, Houston-2018, while the proposed method could achieve the average OA of 71.8% on the target data set with only 5% of the training samples on the source data set.

D. Visualizing the Shared Abundance Space

As discussed in Sections III and IV, with the proposed transfer learning scheme, we are able to project the data from both the source and target domains into a shared abundance space, such that the classifier trained on the source domain can work on the target domain without data labeling or network retraining. This is mainly because, with proper physical constraints, the domain discrepancy has been minimized to a large extent in the abundance space. To further demonstrate the effectiveness of the proposed transfer learning scheme, we project the HSI and its features on the abundance space into a 2-D space using the SVD method for visualization purposes. The results of the three data set pairs are illustrated in Figs. 16–18, respectively. Note that, different colors represent different classes. The class samples on the source domain are drawn with circles, while the class samples on the target domain are drawn with triangles.

We observe that there is some overlap between the spectra of the same class on the source and target domains on the PaviaU-PaviaC data sets, as shown in Fig. 16(a). This explains why some state-of-the-art methods trained on the source domain are able to perform well on the target domain. We further observe that on the second and third data set pairs (images are taken at different time or locations), there exists a large margin between the same class on the source and target domains, as shown in Figs. 17(a) and 18(a). Such discrepancies lead to

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TABLE V

PERFORMANCE COMPARISON WHEN THE MODELS ARE TRAINED AND TESTED ON THE SAME DATA SET, PU

	Method	1D-CNN	Spec-Spat	3D-CNN	HSI-CNN	SSRN	HybridSN	HT-CNN	Ours
5%	OA	98.43±0.42	98.98±0.26	97.3±1.57	97.64±0.23	99.93±0.05	99.68±0.15	99.61±0.19	99.83±0.06
	AA	97.92±0.88	98.86±0.25	97.01±1.54	96.94±0.36	99.95±0.03	99.65±0.18	99.60±0.18	99.85±0.06
	kappa	97.78±0.59	98.56±0.37	96.19±2.22	96.67±0.32	99.90±0.07	99.55±0.22	99.44±0.26	99.76±0.09
3%	OA	97.94±0.13	98.06±0.25	87.59±10.28	95.07±3.1	99.71±0.32	99.55±0.15	99.33±0.35	99.61±0.10
	AA	96.66±0.65	97.76±0.81	88.17±8.70	94.13±2.25	99.72±0.27	99.51±0.11	99.30±0.38	99.66±0.06
	kappa	97.09±0.18	97.26±0.35	83.01±13.85	93.11±4.22	99.59±0.45	99.37±0.21	99.05±0.49	99.44±0.15
1%	OA	97.07±0.75	95.61±0.87	88.5±2.13	93.2±0.96	99.77±0.1	99.38±0.15	97.94±0.50	99.57±0.08
	AA	95.22±1.71	94.86±0.91	78.69±4.40	91.07±0.94	99.77±0.12	99.24±0.17	97.59±0.69	99.55±0.09
	kappa	95.06±1.07	93.80±1.24	83.50±3.11	90.39±1.36	99.68±0.15	99.13±0.21	97.08±0.70	99.39±0.12

TABLE VI

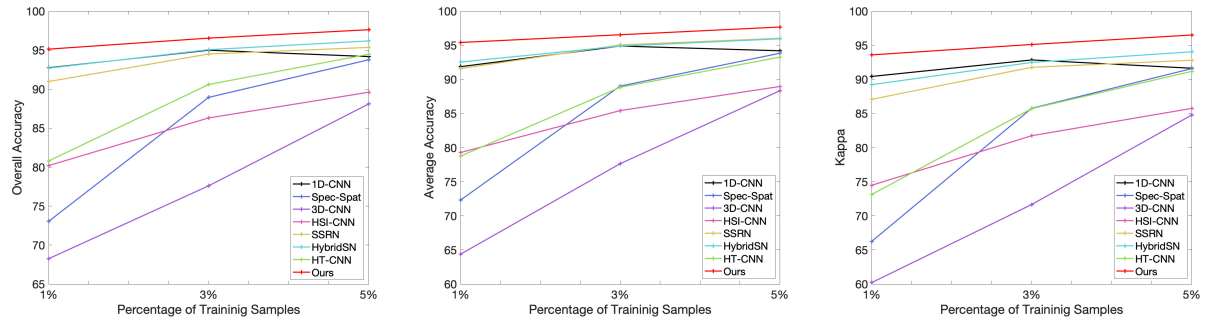
PERFORMANCE COMPARISON WHEN THE MODELS ARE TRAINED AND TESTED ON THE SAME DATA SET, HOUSTON 2013

	Method	1D-CNN	Spec-Spat	3D-CNN	HSI-CNN	SSRN	HybridSN	HT-CNN	Ours
5%	OA	91.98±3.4	88.24±3.84	73.28±4.58	81.41±2.18	94.62±0.72	96.26±1.25	94.81±1.22	96.58±0.29
	AA	90.87±4.14	88.67±3.44	74.29±4.69	82.21±2.21	95.36±0.62	96.41±1.54	92.58±1.64	96.66±0.31
	kappa	90.61±3.99	86.28±4.47	68.81±5.33	78.33±2.55	93.70±0.84	95.63±1.47	93.86±1.43	96.01±0.34
3%	OA	93.2±1.22	75.27±24.37	52.31±19.57	73.83±2.92	92.71±1.41	94.08±1.3	88.02±2.74	94.19±0.89
	AA	92.76±1.27	76.04±24.17	52.30±20.34	74.15±3.06	93.58±1.52	94.65±1.12	83.85±3.23	94.77±0.63
	kappa	92.05±1.43	71.13±28±50	44.10±23.12	69.41±3.41	91.48±1.65	93.08±1.52	85.79±3.25	93.21±1.04
1%	OA	84.06±8.49	31.74±23.02	25.6±8.39	57.37±6.01	85.04±2.94	89.48±3.92	62.52±5.89	89.18±2.03
	AA	82.57±8.93	30.43±24.13	24.45±6.11	57.81±6.32	86.07±3.63	90.00±3.98	59.6±5.75	89.78±1.48
	kappa	81.32±9.93	18.94±27.73	13.14±7.63	50.14±7.04	82.49±3.42	87.68±4.60	55.58±6.93	87.32±2.39

TABLE VII

PERFORMANCE COMPARISON WHEN THE MODELS ARE TRAINED AND TESTED ON THE SAME DATA SET, HANGZHOU

	Method	1D-CNN	Spec-Spat	3D-CNN	HSI-CNN	SSRN	HybridSN	HT-CNN	Ours
5%	OA	92.13±0.64	94.08±0.35	93.74±0.31	89.76±0.52	91.51±0.32	92.62±0.26	89.02±0.77	96.46±0.36
	AA	93.80±0.48	93.97±0.10	93.71±0.49	87.73±0.54	92.79±0.83	91.79±0.20	87.63±1.46	96.53±0.67
	kappa	86.53±1.07	89.91±0.55	89.34±0.51	82.26±0.91	84.82±0.64	86.95±0.43	80.25±1.52	93.76±0.64
3%	OA	93.85±0.75	93.51±0.25	92.91±0.18	90.03±0.83	91.11±0.44	91.56±0.3	84.45±5.33	95.84±0.53
	AA	95.25±0.58	93.25±0.24	92.46±0.48	88.01±1.09	91.86±1.02	90.49±0.55	83.38±4.54	95.21±1.66
	kappa	89.42±1.26	88.90±0.43	87.86±0.34	82.71±1.67	84.25±0.71	85.02±0.57	72.31±9.02	92.71±0.90
1%	OA	97.09±0.8	91.78±1.47	90.66±1.12	90.02±0.69	88.12±1.22	89.22±0.38	81.90±0.65	96.65±0.31
	AA	97.81±0.80	91.64±0.35	90.00±0.74	88.98±0.25	88.92±1.74	88.41±0.59	79.08±0.96	96.91± 0.30
	kappa	94.92±1.39	86.00±2.26	83.99±1.63	82.91±0.98	79.07±1.69	80.88±0.76	66.73±1.28	94.08±0.57



(a) (b) (c)

Fig. 11. Average classification metrics over the three source data sets. (a) Average OA. (b) Average AA. (c) Average kappa.

TABLE VIII

PERFORMANCE COMPARISON WHEN THE MODELS ARE TRAINED ON THE SOURCE DATA SET, PAVIAU, AND TESTED ON THE TARGET DATA SET, PAVIAC

	Method	1D-CNN	Spec-Spat	3D-CNN	HSI-CNN	SSRN	HybridSN	HT-CNN	Ours
5%	OA	83.52±6.37	93.45±0.33	92.96±1.55	93.76±0.87	61.91±4.25	19.08±3.61	79.59±2.05	93.92±0.58
	AA	71.66±7.39	91.56±0.66	90.12±4.34	92.20±1.06	66.41±6.76	17.77±3.37	68.59±0.94	91.78±0.74
	kappa	77.44±8.38	91.02±0.45	90.31±2.21	90.12±1.17	49.20±5.90	-8.70±6.59	71.22±2.79	91.55±0.75
3%	OA	88.6±0.13	93.03±0.47	90.05±2.27	90.79±0.84	66.17±8.09	24.43±1.65	85.19±0.77	93.82±1.21
	AA	77.70±2.13	90.99±2.22	83.74±8.17	88.81±3.64	72.83±5.71	23.20±2.39	75.12±0.99	90.83±1.48
	kappa	84.22±2.26	90.44±0.68	86.16±3.39	87.40±1.25	54.89±11.15	-2.64±3.38	79.11±1.11	91.54±1.63
1%	OA	90.45±0.72	91.04±0.78	87.95±3.21	88.88±0.77	74.9±4.01	17.11±4.98	81.71±3.96	91.15±1.16
	AA	80.00±2.12	90.00±0.97	88.97±2.69	86.93±1.28	70.12±6.01	16.47±4.85	73.18±4.88	87.19±1.78
	kappa	86.78±1.01	87.79±1.05	83.82±4.16	84.86±1.01	66.05±4.62	10.80±6.33	74.35±5.60	87.87±1.61

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TABLE IX

PERFORMANCE COMPARISON WHEN THE MODELS ARE TRAINED ON THE SOURCE DATA SET, HOUSTON 2013, AND TESTED ON THE TARGET DATA SET, HOUSTON 2018

	Method	1D-CNN	Spec-Spat	3D-CNN	HSI-CNN	SSRN	HybridSN	HT-CNN	Ours
5%	OA	39.96±4.76	63.06±7.97	49.37±12,12	53.53±5.69	57.44±2.33	49.41±6.47	57.71±4.61	71.80±3.31
	AA	41.89±6.83	51.02±5.18	53.06±1.25	56.81±0.98	61.41±7.27	41.05±4.62	46.93±2.37	57.80±4.95
	kappa	19.81±5.00	38.97±4.25	31.98±7.37	37.78±5.74	44.07±2.32	27.33±4.87	40.33±3.40	56.36±4.69
3%	OA	48.75±4.4	59.67±5.42	43.73±15.21	50.64±7.03	48.4±7.09	43.13±7.49	58.64±10.20	64.85±2.37
	AA	50.13±6.04	46.68±11.98	40.93±11.62	56.32±2.80	62.55±9.42	43.46±3.70	45.55±3.86	60.00±2.60
	kappa	29.06±4.70	35.66±9.46	23.36±11.25	34.39±5.14	35.50±5.15	24.50±4.78	42.10±8.56	50.15±3.45
1%	OA	67.13±5.61	51.32±11.78	29.88±28.18	55.4±15.02	46.61±8.14	33.19±6.49	34.38±14.7	60.56±4.63
	AA	65.13±7.68	26.81±18.55	21.88±6.14	47.10±5.99	56.29±8.31	39.78±4.19	32.08±11.41	48.10±4.62
	kappa	46.90±4.32	13.56±16.32	8.54±8.23	33.02±10.83	31.56±5.41	17.85±4.91	20.65±8.38	43.33±3.36

TABLE X

PERFORMANCE COMPARISON WHEN THE MODELS ARE TRAINED ON THE SOURCE DATA SET, HANGZHOU, AND TESTED ON THE TARGET DATA SET, SHANGHAI

	Method	1D-CNN	Spec-Spat	3D-CNN	HSI-CNN	SSRN	HybridSN	HT-CNN	Ours
5%	OA	47.61±6.35	67.45±1.48	65.17±2.23	69.28±1.85	66.63±7.56	86.62±3.46	77.71±5.87	90.01±2.43
	AA	54.30±0.98	73.59±1.22	71.68±2.04	75.13±1.14	78.17±3.26	88.02±3.35	78.29±7.07	90.01±1.82
	kappa	26.86±9.35	53.89±1.97	50.83±3.14	56.37±2.39	46.71±12.64	79.37±5.42	66.33±8.85	84.65±3.77
3%	OA	50.08±5.66	67.73±1.62	63.28±4.09	67.57±4.05	79.47±5.65	88.81±2.01	70.99±13.45	89.25±2.29
	AA	56.08±1.55	73.89±1.28	70.30±3.84	73.85±3.10	83.68±2.45	89.91±2.02	73.85±13.27	88.96±1.90
	kappa	30.11±8.09	54.33±2.11	48.50±5.83	54.24±5.21	67.98±9.16	82.75±3.16	57.47±19.22	83.69±3.41
1%	OA	48.06±10.15	67.75±3.78	66.93±3.82	62.63±3.22	66.55±7.06	88.1±1.71	68.54± 0.65	88.61±2.52
	AA	57.79±0.74	73.97±2.76	73.62±2.94	70.11±2.53	69.14±15.76	88.72±1.93	74.44±0.75	88.50±2.12
	kappa	26.39±15.58	54.46±4.71	53.64±4.80	48.02±4.06	46.70±12.05	81.63±2.69	54.95±0.97	82.71± 3.70

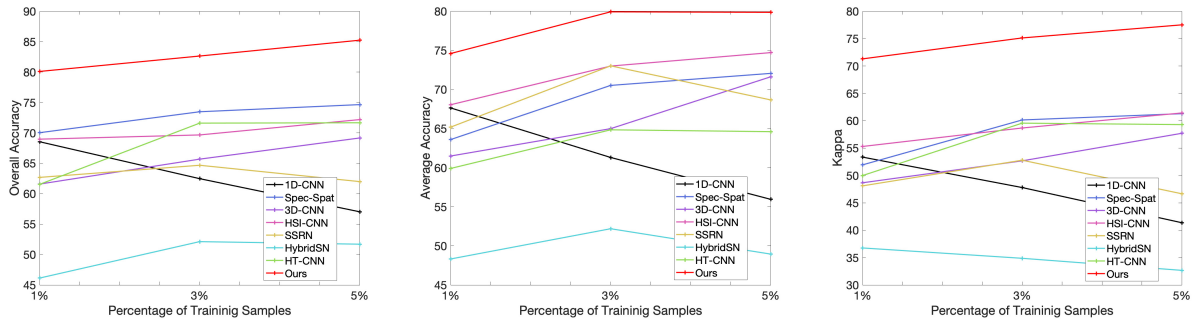


Fig. 12. Average classification metric over the three target data sets. (a) Average OA. (b) Average AA. (c) Average kappa.

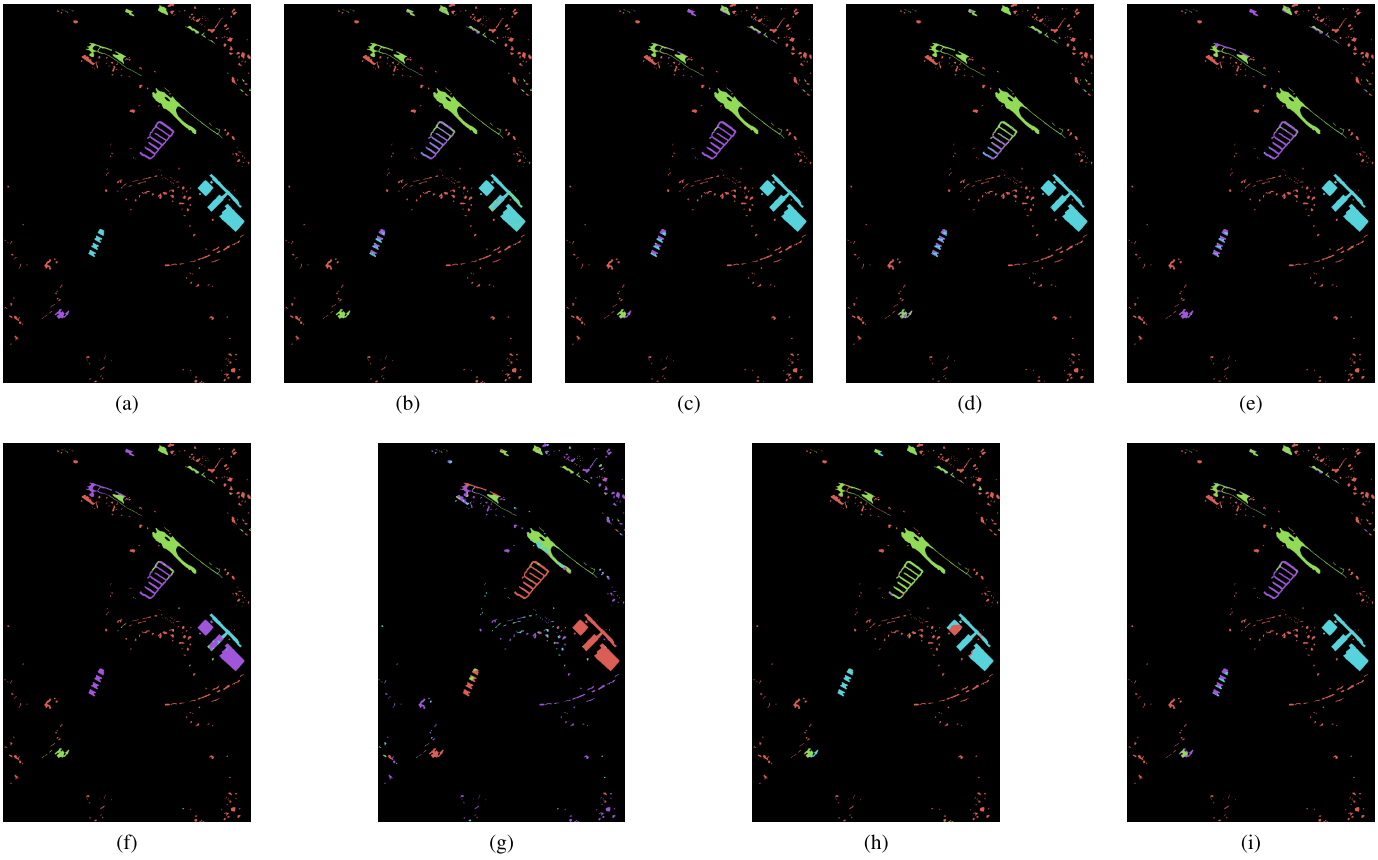
the failure of the state-of-the-art classifiers. On the other hand, regardless of the type of data sets used, the proposed methods can always minimize the domain discrepancy between the source and target data sets by projecting the data samples onto the shared abundance space with physical constrains, as shown in Figs. 16(b), 17(b), and 18(b), where more samples of the same classes overlap although they are from different domains.

E. Ablation Study

In this section, we would like to evaluate the contributions of the important components of the proposed scheme. As shown in Fig. 2, the architecture of the proposed PCTL-SAS consists of two branches, i.e., 1) a reconstruction branch with shared Dirichlet-encoder, an affine-transfer decoder, and a mutual discriminative network designed based on the physical constraints and 2) a classification branch with densely connected 3-D-CNN classifier. The reconstruction branch projects both the source and target images onto the shared abundance space, and the classifier is concatenated to the shared abundance space, such that even though the network is trained on the source domain, it could still characterize the target domain without additional data labeling or retraining. In this section, we provide comprehensive ablation studies of the above components to discuss the effectiveness of the proposed method. The first data set is used to conduct the ablation studies and the average OA is adopted to evaluate the classification performance.

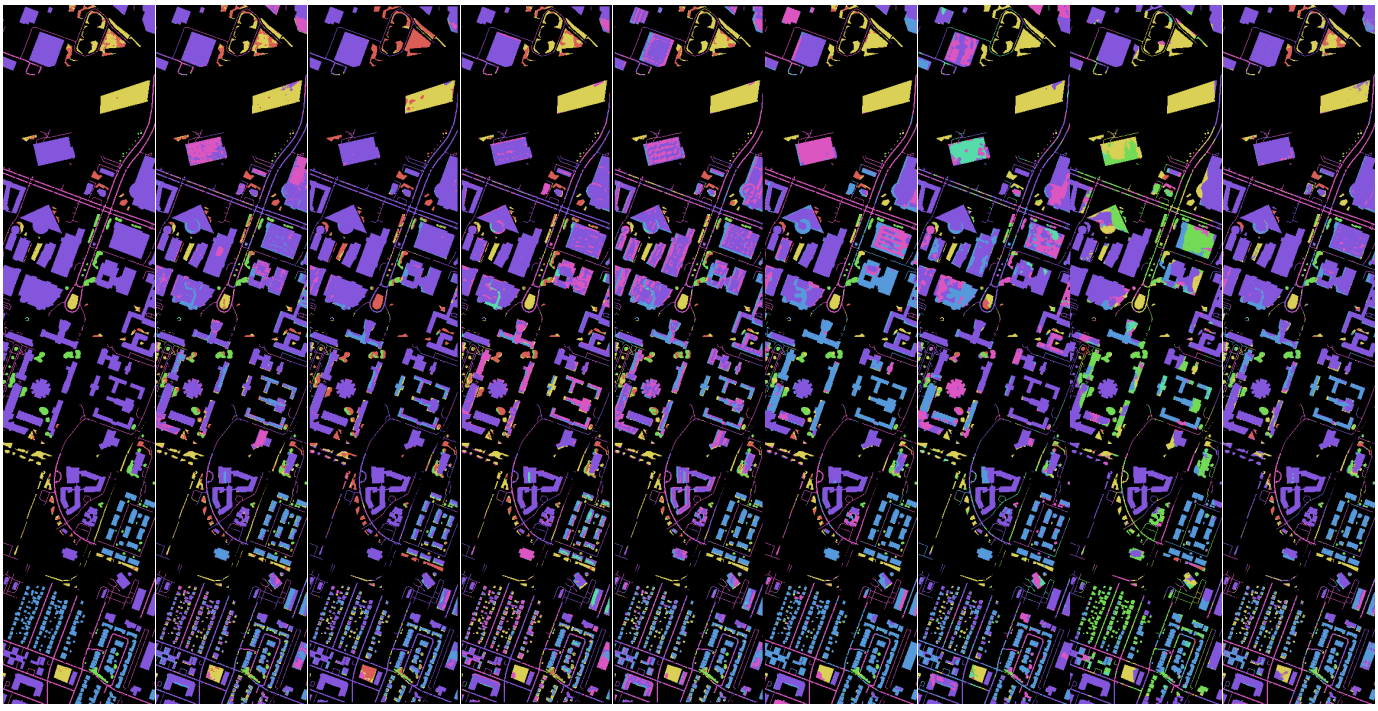
1) Effects of the Classifier: We first conduct two experiments to evaluate the densely connected 3-D-CNN classifier without the reconstruction branch. The results of the two experiments are reported in Figs. 19 and 20, respectively. From Fig. 19, we can observe that if only the 3-D classifier is adopted, its accuracy on the target data set drops significantly. That is because the classifier itself could not extract the shared intrinsic representations between images in different domains. Therefore, the extraction of the shared representations would contribute significantly to the classification performance in

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(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 13. (a) Ground truth classification map of the target image, PC. The predicted map from (b) 1-D-CNN, (c) Spec-Spat, (d) 3-D-CNN, (e) HSI-CNN, (f) SSRN, (g) HybirdSN, (h) HT-CNN, and (i) Ours.



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 14. (a) Ground truth of the target image, houston 2018. The predicted classification map from (b) 1-D-CNN, (c) Spec-Spat, (d) 3-D-CNN, (e) HSI-CNN, (f) SSRN, (g) HybirdSN, (h) HT-CNN, and (i) Ours.

a different domain. In the second experiment, besides the 1% labeled source samples, we add additional 0.5% labeled target samples during the training procedure to evaluate the performance of the classifiers on the three data set pairs. The OA of the classifiers is demonstrated in Fig. 20. We can observe that the proposed 3-D-CNN classifier could achieve

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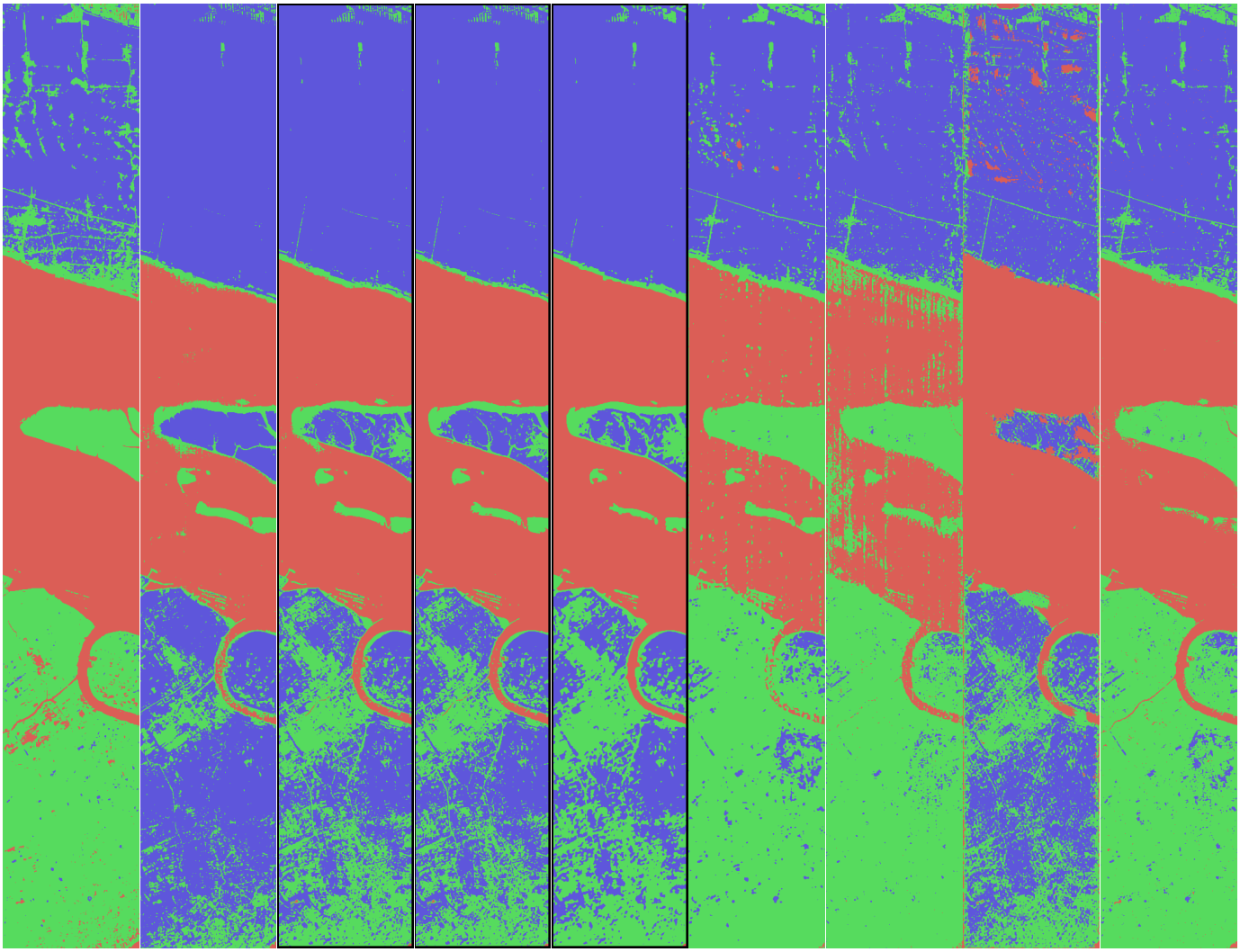


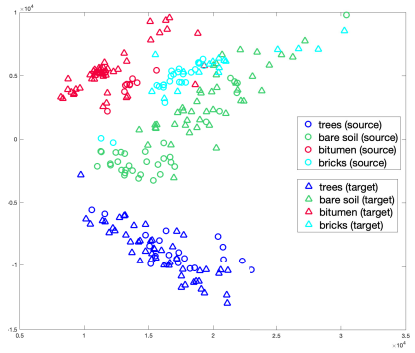
Fig. 15. (a) Ground truth of the target image, Shanghai. The predicted classification map from (b) 1-D-CNN, (c) Spec-Spat, (d) 3-D-CNN, (e) HSI-CNN, (f) SSRN, (g) HybirdSN, (h) HT-CNN, and (i) Ours.

better performance as compared to the one trained only on the 1% source samples, since it is able to extract effective representations from the target domain with 0.5% labeled target samples. With the reconstruction branch, the proposed method is able to extract effective shared intrinsic representations, which further increase the performance by 3.1%–6.8% on the three data sets.

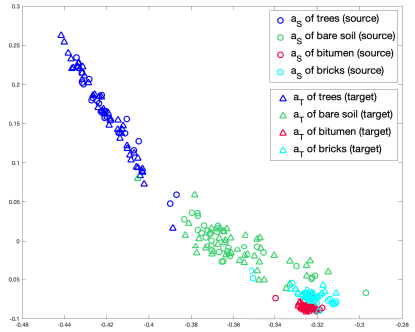
2) Effects of the Affine-Transfer Decoder: The affine-transfer decoder, designed based on the potential physical relationship between the same-class samples in different domains, is an important component of the reconstruction branch which allows the network to project the data from both the source and target domains into a shared abundance space. In the second experiment, we evaluate the effect of the affine-transfer decoder and the results are demonstrated in Fig. 19 as well. Assume that there is only one general decoder to reconstruct both the source image and target images. The classification performance on the target image increases at least 3% percent because the network learns the representations from both domains. When we adopt the proposed affine-transfer decoder, the accuracy on the target domain is further increased. This is because the network is enforced to simulate the potential physical relationship between different domains, which allows the network to extract shared representations between the source and target images.

3) Effects of the Sparse Constraint and Mutual Discriminative Network: We also observe from Fig. 19 that the proposed method with more physical constraints based on sparse regularization and mutual discriminative network could increase the OA of the target image by a large margin. In this study, we evaluate how these two constraints influence the transfer learning ability of the model on the target image. We first fix the parameter, λ, for mutual discriminative network, and adjust the parameter, α, for sparse constraint. The result is shown in Fig. 21(a). We can observe that, when we increase α, the average OA gradually increases in most scenarios, which means the sparse-regularized with entropy function is able to enforce the network to extract more effective representations. But when α is too large, the performance is decreased because it reduces the reconstruction accuracy which, in turn, prevents the network to extract representative features. Thus, in our network settings, $\alpha = 0.001$.

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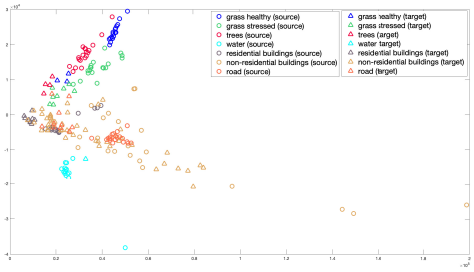


(a)

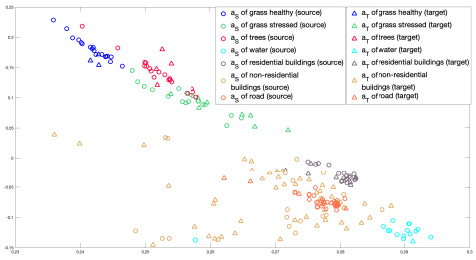


(b)

Fig. 16. (a) Spectra and (b) abundance vector of the classes on the source domain, PaviaU, and the target domain, PaviaC. Different colors represent different classes. Circles indicate the class samples on the source domain, and triangles indicate the class samples on the target domain.



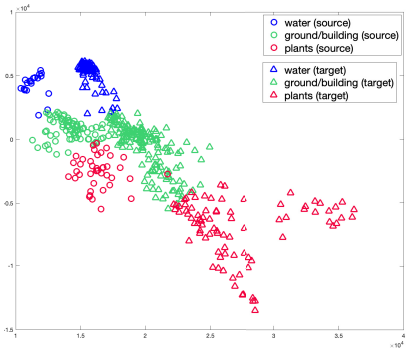
(a)



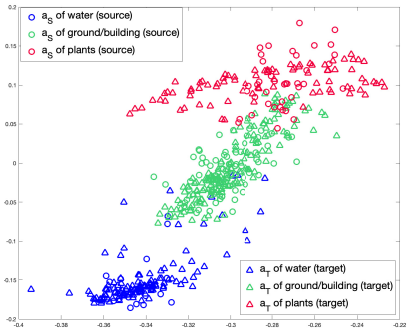
(b)

Fig. 17. (a) Spectra and (b) abundance vector of the classes on the source domain, Houston-2013, and the target domain, Houston-2018. Different colors represent different classes. Circles indicate the class samples on the source domain, and triangles indicate the class samples on the target domain.

In Fig. 21(b), we fix $\alpha = 0.001$ and adjust the parameter λ. We can observe that in most scenarios when λ increases, the average OA becomes better. The main reason is that the mutual discriminative network enforces the correspondence between the pixels belonging to the same classes in different domains better. Thus, in our experimental settings, $\lambda \in \{0.01, 0.1\}$.



(a)



(b)

Fig. 18. (a) Spectra and (b) abundance vector of the classes on the source domain, Hangzhou, and the target domain, Shanghai. Different colors represent different classes. Circles indicate the class samples on the source domain, and triangles indicate the class samples on the target domain.

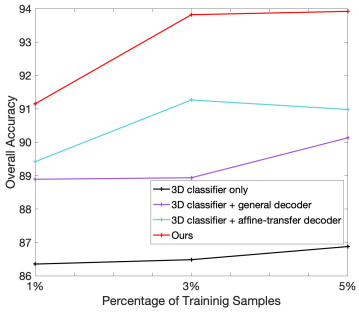


Fig. 19. Average OA of the trained model when tested on the target data set.

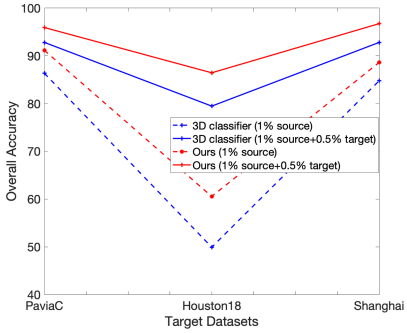
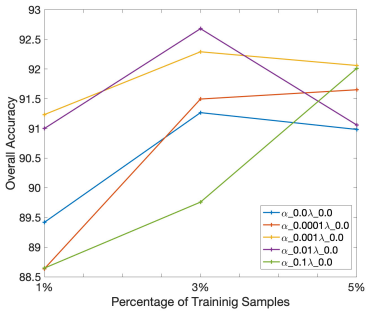
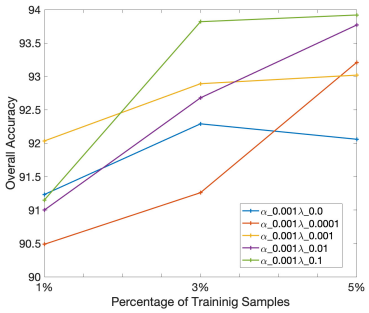


Fig. 20. Average OA of the classifiers when trained with both source and target data set and tested on the target data set.

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(a)



(b)

Fig. 21. Average OA on the target domain with parameter, λ for the mutual discriminative network and α for the sparse constraint. (a) Fix λ and adjust α. (b) Fix α and adjust λ.

VI. Conclusion

We presented a new transfer learning scheme for HSI classification such that a trained model from the source domain does not need to be retrained or fine-tuned using extra labeled data from the target domain. The key to this new learning scheme is the construction of a shared abundance space that would largely reduce the discrepancies between the different domains, due to the intrinsic physical constraints it must incorporates when constructing such a space. The learning scheme includes three novel developments. First, a shared Dirichlet-encoder is proposed that is constrained by the sparse entropy function to extract effective representations on the shared abundance space from the raw data of different domains. Second, an affine-transfer decoder regularized by the mutual discriminative network is proposed to extract shared intrinsic spectral bases such that pixels of the same class but from different domains can be matched. Third, a densely connected 3-D-CNN classifier is proposed and concatenated to the shared encoder, such that the classifier could be trained on the source domain and perform on the target domain without extra efforts for data labeling or network retraining. Extensive experiments on different types of data pairs demonstrates the superiority of the proposed approach over the state of the art.

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Ying Qu (Member, IEEE) received the B.S. degree in automatics and the M.S. degree in pattern recognition and artificial intelligence from Northeastern University, Shenyang, China, in 2008 and 2010, respectively. She is pursuing the Ph.D. degree in computer engineering with the University of Tennessee, Knoxville, TN, USA.

Dr. Qu was a recipient of the Best Student Paper Awards at The International Geoscience and Remote Sensing Symposium (IGARSS) in 2016.



Razieh Kaviani Baghbaderani (Student Member, IEEE) received the B.S. degree in electrical engineering from the Isfahan University of Technology, Isfahan, Iran, in 2012, and the M.S. degree in electrical engineering from the Sharif University of Technology, Tehran, Iran, in 2014. She is pursuing the Ph.D. degree in electrical engineering with the University of Tennessee, Knoxville, TN, USA.

Her research interests include computer vision and hyperspectral image analysis.

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Wei Li (Senior Member, IEEE) received the B.E. degree in telecommunications engineering from Xidian University, Xi'an, China, in 2007, the M.S. degree in information science and technology from Sun Yat-sen University, Guangzhou, China, in 2009, and the Ph.D. degree in electrical and computer engineering from Mississippi State University, Starkville, MS, USA, in 2012.

Subsequently, he spent one year as a Post-Doctoral Researcher with the University of California, Davis, CA, USA. He was a Professor with the College of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China from 2013 to 2019. He is a Professor with the School of Information and Electronics, Beijing Institute of Technology, Beijing. His research interests include hyperspectral image analysis, pattern recognition, and data compression.

Dr. Li received the 2015 Best Reviewer Award from the IEEE Geoscience and Remote Sensing Society (GRSS) for his service for IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING (JSTARS) and the Outstanding Paper Award at IEEE International Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensing (Whispers) in 2019. He is serving as an Associate Editor for the IEEE SIGNAL PROCESSING LETTERS and the IEEE JSTARS. He has served as a Guest Editor for special issue of *Journal of Real-Time Image Processing*, *Remote Sensing*, and IEEE JSTARS.



Lianru Gao (Senior Member, IEEE) received the B.S. degree in civil engineering from Tsinghua University, Beijing, China, in 2002, and the Ph.D. degree in cartography and geographic information system from the Institute of Remote Sensing Applications, Chinese Academy of Sciences (CAS), Beijing, in 2007.

He is a Professor with the Key Laboratory of Digital Earth Science, Aerospace Information Research Institute, CAS. He also has been a Visiting Scholar with the University of Extremadura, Cáceres, Spain, in 2014, and the Mississippi State University (MSU), Starkville, MS, USA, in 2016. His research focuses on hyperspectral image processing and information extraction. In last ten years, he was the principle investigator (PI) of ten scientific research projects at national and ministerial levels, including projects by the National Natural Science Foundation of China from 2010 to 2012, from 2016 to 2019, from 2018 to 2020, and by the Key Research Program of the CAS from 2013 to 2015 and so on. He has published more than 160 peer-reviewed articles, and there are more than 80 journal articles included by Science Citation Index (SCI). He has coauthored an academic book entitled *Hyperspectral Image Classification and Target Detection*. He received 29 National Invention Patents in China. He was received the Outstanding Science and Technology Achievement Prize of the CAS in 2016, and was supported by the China National Science Fund for Excellent Young Scholars in 2017, and received the Second Prize of The State Scientific and Technological Progress Award in 2018. He received the recognition of the Best Reviewers of the IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING in 2015, and the Best Reviewers of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING in 2017.



Yuxiang Zhang (Student Member, IEEE) received the B.S. degree in electronics and information engineering and the M.S. degree in control science and engineering from the Beijing University of Chemical Technology, Beijing, China, in 2017 and 2020, respectively. He is pursuing the Ph.D. degree with the Beijing Institute of Technology, Beijing.

His research interests include hyperspectral imagery processing and pattern recognition. His supervisor is Dr. Wei Li.



Hairong Qi (Fellow, IEEE) received the B.S. and M.S. degrees in computer science from Northern Jiaotong University, Beijing, China, in 1992 and 1995, respectively, and the Ph.D. degree in computer engineering from North Carolina State University, Raleigh, NC, USA, in 1999.

She is the Gonzalez Family Professor with the Department of Electrical Engineering and Computer Science, University of Tennessee, Knoxville, TN, USA. Her research interests are in advanced imaging and collaborative processing in resource-constrained distributed environment, hyperspectral image analysis, and automatic target recognition. Her research was supported by the National Science Foundation (NSF), Defense Advanced Research Projects Agency (DARPA), Office of Naval Research (ONR), Department of Homeland Security (DHS), U.S. Army Space and Missile Defense Command, and U.S. Army Medical Research and Materiel Command.

Dr. Qi was a recipient of the NSF CAREER Award. She also received the Best Paper Awards at the 18th International Conference on Pattern Recognition (ICPR) in 2006, the 3rd ACM/IEEE International Conference on Distributed Smart Cameras (ICDSC) in 2009, and the IEEE Workshop on Hyperspectral Image and Signal Processing: Evolution in Remote Sensor (WHISPERS) in 2015. She was received the Highest Impact Paper from the IEEE Geoscience and Remote Sensing Society in 2012.

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